

Week 2 Assignment: Clean It, Code It, Frame It

Aaron Foster

Southern New Hampshire University

CS-540: Data Literacy & Visualization

Prof. Mahmoud Al-Khassaweneh

May 3rd, 2026

The Value of Structured Data in Business Analytics

Transforming raw, messy data into a clean tabular format is the foundation of meaningful analysis and informed decision making. In my field of interest, business analytics, decisions about pricing, staffing, inventory, and customer strategy all depend on the integrity of the underlying data. A single unaddressed flaw, a duplicate transaction, a sentinel value mistaken for a real measurement, or a date stored as text which can quietly distort an entire analysis and lead to costly choices.

This assignment really shows how important good structure is when working with data. The original weather dataset had a lot of issues—284 duplicate rows, strange placeholder values like 5505 and -40 pretending to be real temperatures, and even infinity values that could break calculations. If those problems weren't fixed, any summaries, charts, or models built from the data would end up being unreliable or just plain wrong. By cleaning things up—replacing those placeholder values with NaN, removing duplicates, converting the date column into a proper datetime format, and making the column names consistent—the dataset becomes much more dependable. After that, it's in a solid state to be used with tools like Pandas, SQL, or visualization platforms such as Tableau and Power BI.

A clean tabular format also supports collaboration and communication. When data is organized into consistent columns with correct data types, analysts, managers, and stakeholders can interpret it the same way without ambiguity. Filters, joins, pivot tables, and dashboards all assume tabular structure, so clean data accelerates every stage of the analytics pipeline. Ultimately, transforming raw data into a structured table is not just a technical exercise, it is the

step that converts noise into evidence. In business analytics, that evidence drives the strategic decisions that determine whether an organization grows, stalls, or fails. Reliable data leads to reliable decisions, and reliable decisions are the foundation of competitive performance.